

LensJudge: An Agentic Visual-Inspection Grader for Strong-Lens Candidates on the Claude Agent SDK

Greg Benson

Xiaosheng Huang

June 2026

Internal Prototype + LensBench-VI Evaluation

Abstract

The Huang group’s ResNet/EfficientNet finders emit thousands of above-threshold cutouts per data release; turning them into graded candidates (A/B/C/D) is a human visual-inspection (VI) bottleneck. We build **LensJudge** — the system the group’s onboarding report calls “LensAgent”/“VI Pre-grading Agent”, renamed to avoid a name clash with an existing arXiv paper — on the *Claude Agent SDK for Python*, and ask whether the data on hand can *test* such a system. The answer is yes for imaging: $\sim 2,706$ human-graded A/B/C cutouts, $\sim 2,345$ Grade-D human-*rejected* hard negatives, $\sim 65\text{K}$ random-galaxy negatives, and a Foundry-II confirmed/non-lens gold tier are all in hand. We implement three graders — a lean single multimodal call, a perspective-diverse judge panel, and a factor-decomposition multi-agent design — plus a spectroscopic pre-grader for discordant-redshift pairs, and score them in a shared harness (LensBench-VI) on a frozen held-out set of 200 imaging systems. Because every label is a single 2-author consensus (no per-rater data), all agreement numbers are *consensus-referenced with no human ceiling*. Two results stand out. (1) On the *spectroscopic* task — clean physical features, spectroscopically confirmed gold — the agent is strong: 20/20 Hsu Table-2 Grade-A pairs re-graded plausible, 4/4 Foundry-II non-lenses rejected. (2) On *imaging* grading of the CNN’s hard high-score pool, no configuration — lean Sonnet, lean Opus, a judge panel, or the $\$9.1$ multi-agent design — reproduces the single-consensus A/B/C grades: all four are near-chance (AUC 0.32–0.51) and uniformly skeptical, and paying $2\text{--}4\times$ more per candidate buys nothing. We argue this is less a model failure than a measurement problem — A/B/C are mostly *unconfirmed* candidates and the C/D boundary is one rater’s subjective call — which turns the missing multi-grader ceiling from a caveat into the central finding.

1 Motivation

Lens finding in wide-area imaging is a two-stage pipeline: a CNN ranks $O(10^7)$ galaxy cutouts, and humans visually grade the $O(10^3\text{--}10^4)$ above-threshold survivors into A (clear lens), B (probable), C (possible), and D (rejected) (Huang et al., 2020, 2021; Inchausti Reyes et al., 2025; Storfer et al., 2024). The VI step is the wall-clock bottleneck and will only worsen at Rubin scale. LensJudge automates the structured judgment of that step with a tool-using LLM agent, pre-grading candidates and escalating the genuinely ambiguous ones to a human — it is a triage assistant, not an autonomous arbiter.

2 Do we have enough data to test it?

Imaging: yes, today. From the group’s reproductions we assemble (all public / on-disk): *positives* — 2,706 graded A/B/C candidates as $101 \times 101 \times 3$ *grz* FITS (Storfer DR9 + Inchausti DR10) with per-candidate CNN probabilities; *hard negatives* — 2,345 Grade-D human-rejected candidates (the false positives the CNN flagged but humans rejected) parsed from the published catalogs, cutouts fetched on demand; *easy negatives* — ~ 65 K random-galaxy cutouts; *gold* — 21 Foundry-II confirmed/known lenses + 4 confirmed non-lenses (Huang et al., 2025). **Spectroscopy: catalog-level today.** Hsu’s 13,530 discordant-redshift fiber pairs and 20 Table-2 Grade-A systems (Hsu et al., 2025) give a spectroscopic test set, though the raw DESI fiber flux is not local.

Two honest gaps — one now partly closed externally. (1) Every *DESI* grade is a *single* 2-senior-author consensus — there are no per-rater labels in the DESI catalogs — so a human-vs-human Cohen- κ ceiling cannot be computed from them. We later locate external data that re-anchors a subset (§10): 291 candidates carry higher-resolution imaging, multi-grader grades, or spectroscopic confirmation, and we stand up the Euclid Q1 ($0.1''$, ~ 10 -vote) field as a high-resolution benchmark. A per-rater κ over the *DESI* grades still needs the ~ 250 -cutout in-house study. (2) The spectroscopic grader reasons over catalog features (z_l, z_s , separation, σ_v, θ_E) plus DR10 imaging; raw-flux reading (SPARCL / public healpix coadds) is a documented enhancement.

3 Architecture on the Claude Agent SDK

The SDK has no image-in-prompt, so the only way the model sees pixels is a tool that returns an image content block. Our `fetch_cutout` tool renders the *grz* cube to multi-view Lupton-RGB PNGs (full / center-zoom / lens-light-residual / high-contrast, identical params to the group’s inspection viewer) and returns them; `get_photometry` measures aperture $g-r, r-z$ colors; `crossmatch_local` flags prior-catalog overlap; `get_specfit` does the SIS θ_E /separation check. Tools are an in-process `create_sdk_mcp_server` bundle; because the Python `@tool structuredContent` is not forwarded, the grade contract is a system-prompt JSON schema validated into Pydantic models with one repair retry; PreToolUse/PostToolUse hooks stream a per-candidate JSONL trace (with `total_cost_usd`) that doubles as the eval harness.

We implement three imaging graders of increasing cost: **lean** (one multimodal call; the five Huang-2020 criteria as scored fields of a single integrated judgment), **panel** (four perspective-diverse judges — Advocate, Skeptic, Morphologist, Contaminant-hunter — fused by ordinal quorum + skeptic-veto + a 2-of-N A-rule), and **multiagent** (the onboarding-plan §9.1 factor-decomposition: morphology/color/ contaminant specialists $\rightarrow$ GradeArbitrator). A **spectro** grader classifies discordant-redshift pairs lens/dimple/not.lens.

4 LensBench-VI methodology

The held-out set is a frozen, hash-pinned manifest of 200 systems: graded A/B/C positives + Grade-D hard negatives + random-galaxy negatives + Foundry-II gold, stratified by grade and region, negatives kept as separate strata. The LLM grader never trained on any cutout, so all rows are valid held-out for it (the NeuraLens training-set leakage that affects the frozen-CNN baseline is recorded per row). Tasks/metrics: (i) binary lens(A/B/C) vs non-lens(D) — ROC-AUC and recovery at 1%/0.1% FPR (the group’s operating point); (ii) ordinal A/B/C/D vs consensus — quadratic-weighted κ , exact/adjacent accuracy, confusion; (iii) calibration of p_{lens} (ECE, Brier);

(iv) agent-vs-CNN agreement. Baselines: the p_{meta} threshold, single nets, majority-class, and image-only-no-tools.

5 Results

On the full 200-system clean-imaging held-out set the lean Sonnet grader scores binary ROC-AUC 0.51, recovery@1% FPR 0.04, quadratic-weighted $\kappa = -0.02$, ECE 0.42 at \$0.06/candidate — i.e. essentially no agreement with the single-consensus A/B/C labels. Table 1 asks whether a stronger backbone or multi-agent deliberation recovers it, on a shared 48-system slice. All numbers are **consensus-referenced, no human ceiling**.

Table 1: Grader comparison on the shared 48-system imaging slice (consensus-referenced; no human ceiling). No configuration separates from chance.

Metric	lean-Sonnet	lean-Opus	panel	multiagent
Binary ROC-AUC	0.44	0.32	0.42	0.38
Quad-weighted κ	-0.13	-0.06	-0.07	-0.13
Mean cost / candidate (\$)	0.06	0.13	0.25	0.20

Key finding — on the hard imaging pool, neither the agent nor the CNN separates, and more compute does not help. The frozen ensemble cannot separate real lenses from the Grade-D human-rejects (mean $p_{\text{meta}} \approx 0.82$ on rejects, near its value on lenses) — but, crucially, *nor does the agent*: every configuration sits at AUC 0.32–0.51 and $\kappa \approx 0$, assigning $p_{\text{lens}} \approx 0.02$ –0.03 even to consensus-A systems, and a 2–4 $\times$ more expensive Opus / panel / multi-agent run does not improve it. The agents are *uniformly skeptical* of the whole high- p_{meta} pool.

Why this is the right answer, not a bug. We verified the cutouts are real and correctly resolved (all on disk; obvious arcs, e.g. DESI-091.7214–58.9787, do get high p_{lens}). The pool is the hardest possible slice — positives and CNN-rejected negatives *both* have high p_{meta} — and on it the A/B/C labels are mostly *unconfirmed* candidates whose C-vs-D boundary is one rater’s subjective call. So an agent that is skeptical of unconfirmed C-grade candidates is not demonstrably wrong; it simply cannot be *adjudicated* against a single-consensus label. This is precisely what the missing multi-grader ceiling (and the absence of spectroscopic confirmation for most candidates) prevents us from resolving — the gap is the result. Appendix A shows three of these grading episodes end-to-end — a consensus-A lens, an ambiguous grade-C, and a Grade-D human-reject — abbreviated but otherwise verbatim, to give a concrete feel for how the agent reasons and uses its tools.

6 Reasoning made visible: Sonnet 4.6 vs Opus 4.8

Two instrumentation upgrades motivate this section. First, every grader call now *records the model’s reasoning*: the Agent SDK’s adaptive-thinking summaries are written into the per-candidate trace logs alongside the existing tool events, and the worked examples in Appendix A display them inline. Second, with reasoning observable we re-ran the lean grader on two backbones — `claude-sonnet-4-6` and `claude-opus-4-8` (1M context) — holding the rubric, tools, 6-turn limit and per-candidate budget cap fixed (the cap was raised to \$1.00 for Opus, whose episodes are costlier). Two caveats frame everything below: the displayed text is the API’s *summarized* reasoning, not the raw chain of thought; and instrumenting capture revealed that adaptive thinking is the harness *default* — the frozen Table 1 runs were already reasoning silently. The comparison is therefore “reasoning made visible + backbone swapped”, not “reasoning switched on”.

Table 2 scores all three configurations on the identical 225-row manifest (the frozen baseline is re-scored on the full row set, which is why its numbers differ slightly from the $n=200$ clean-slice values quoted in §5).

Table 2: Backbone comparison with reasoning captured on the full 225-row LensBench-VI manifest (consensus-referenced; no human ceiling). “Baseline” is the frozen Table 1 lean-Sonnet run re-scored on the identical rows; its reasoning was on by default but unrecorded.

Metric	lean-Sonnet (frozen baseline)	Sonnet 4.6 reasoning captured	Opus 4.8 (1M) reasoning captured
Binary ROC-AUC	0.48	0.46	0.44
Quad-weighted κ	-0.08	-0.10	-0.03
Recovery @1% FPR	0.04	0.01	0.01
Calibration ECE	0.46	0.47	0.58
Escalation rate	0.48	0.42	0.12
Mean p_{lens} on truth-A	0.13	0.14	0.05
JSON parse rate	100%	100%	100%
Mean cost / cand. (\$)	0.06	0.08	0.15
Mean wall (s)	43	47	33
Mean thinking chars	—	3,359	1,647

Imaging: the wall is real, reproducible, and backbone-independent. The Sonnet re-run lands within noise of the frozen baseline (AUC 0.46 vs 0.48, κ -0.10 vs -0.08) — the near-chance imaging result of §5 is stable run-to-run, not an artifact of one sample. Opus 4.8, at roughly $2\times$ the per-candidate cost, does not buy agreement with the consensus labels either; it is the most decisively skeptical grader yet. It did not assign a single A grade on the 225-row manifest, graded 26 of the 30 consensus-A systems D, held p_{lens} flat at ~ 0.05 across all four truth grades, and escalated only 0.12 of cases (versus 0.42 for Sonnet) — where Sonnet hedges and hands borderline cases to a human, Opus rejects them with confidence. On the appendix’s consensus-A system (DESI-091.7214–58.9787) the two backbones split exactly where the label is softest: Sonnet graded B/ $p_{\text{lens}}=0.76$ three times running, Opus C/0.12. Curiously, Opus also produces *shorter* summarized reasoning than Sonnet (1,647 vs 3,359 characters per candidate) and finishes faster (33s vs 47s) — its cost premium is price per token, not verbosity.

Spectroscopy: the gold-set result survives, with a twist. With reasoning captured, Sonnet re-grades 19/20 Hsu Table-2 pairs plausible and rejects 4/4 Foundry-II non-lenses (the frozen run scored 20/20 and 4/4); it accepts 17/20 of the Foundry-II *confirmed* lenses. Opus roughly matches it on Hsu (18/20) and on the non-lenses (4/4), but accepts only 14/20 confirmed lenses — six reasoned rejections versus Sonnet’s three. (A first Opus pass lost three episodes to JSON parse failures; we added the imaging path’s one-shot repair retry to the spectro grader, after which all 44 parse and three of the apparent rejections resolved to acceptances.) The remaining rejections are not noise — Opus applies stricter physical-plausibility checks and catches real data-quality gaps, e.g.:

Opus 4.8 — rationale for rejecting a confirmed lens (excerpt)

theta_E is only 0.26" (sigma_v=152 km/s is modest), yet the imaging shows the companion ~3-4" north of the lens --- roughly 10x the Einstein radius, far outside any lensable geometry (the specifit 'strongly lensable' verdict is an artifact of the missing/zeroed separation). [...] Despite the discordant redshifts, this is a chance projection of two galaxies, not a strong lens [...]

On systems whose catalog features are incomplete, that stricter prior errs toward rejection — and on *confirmed* lenses, skepticism is not free. The stronger backbone buys better-argued verdicts, not

better agreement.

For a qualitative feel for the two reasoning styles, compare the Sonnet episodes in Appendix A with this excerpt from the Opus episode on the same consensus-A system (it called `get_photometry`, which none of the three Sonnet recordings did):

Opus 4.8 thinking (summarized) — DESI-091.7214–58.9787, excerpt

The photometry suggests the annulus is bluer than the core, which weakly supports the first criterion, but the morphology is really what I need to focus on. [...] Critically, there’s no blue tangential arc, no ring, and no symmetric counter-image pair. The greenish diffuse light around the core isn’t actually blue and doesn’t have an arc shape, so it doesn’t fit the lensing morphology I’d expect. [...] This sits right at the C/D boundary. The blue excess in the annulus and low-surface-brightness diffuse light near the lens point toward C (possible lens with a blue neighbor at plausible separation but weak morphology) [...] I’ll set `p_lens` to 0.12, confidence to 0.6, leave contaminant null since this doesn’t fit a clear artifact category, and mark it for human eyes given the ambiguity.

6.1 A prompt ablation: absolute evidence vs. relative triage

Opus’s behavior suggests a hypothesis: the rubric states near-absolute evidence standards, and the more literal model applies them as written — while the human graders it is scored against were doing *relative triage* within a pool the CNN had already selected. We test this with a second rubric (`prompts/rubric_imaging_triage.md`) that holds the five criteria, tools, and output schema fixed and changes only the framing: grades are follow-up priority bins relative to the pool at 1.3” seeing; a textual “house scale” describes what consensus A/B/C/D actually look like at this resolution; a D *requires* a named contaminant; and the escalation policy is made asymmetric (“cannot confirm or refute” routes to C + escalate, not to D). p_{lens} is explicitly kept as an *absolute* probability. After a pilot on the shared 48-system slice confirmed the effect, both rubrics ran on the full 225-row manifest, whose strata let us score behavior on *hard* labels too: the 21 Foundry-II spectroscopically confirmed lenses (“gold”) and the 30 random-galaxy negatives.

Table 3: Rubric ablation on the full 225-row manifest. “Original” rows are the Table 2 runs. Recovery_{ABC} = fraction of consensus-A/B/C systems kept in a lens bin; Gold = fraction of the 21 *confirmed* lenses kept in a lens bin (hard labels, not consensus).

Config	Escal.	D-rate	Recovery_{ABC}	Gold	mean p_{lens} (truth-A)	AUC
lean-Sonnet (frozen)	0.48	0.46	0.52	0.38	0.13	0.48
Sonnet 4.6, original	0.42	0.53	0.42	0.33	0.14	0.46
Sonnet 4.6, triage	0.70	0.35	0.66	0.57	0.15	0.51
Opus 4.8, original	0.12	0.88	0.08	0.14	0.05	0.44
Opus 4.8, triage	0.68	0.63	0.40	0.33	0.09	0.49

The split is clean: **the routing policy is prompt-fixable; the discrimination is not.** Under the triage rubric Opus’s operationally dangerous failure mode disappears — escalation recovers from 12% to 68%, the reject-everything D-rate falls from 88% to 63%, and five times as many consensus-graded candidates stay in the lens bins. The gold tier makes the stakes concrete *on hard labels*: with the original rubric Opus kept only 14% of spectroscopically *confirmed* lenses in a lens bin — it would have silently discarded six of every seven proven lenses — versus 33% under triage framing (Sonnet: 33% → 57%). Yet no configuration’s ranking moves outside noise: every AUC sits at chance (0.44–0.51, $n=225$), Opus still assigns essentially no A or B grades, and its absolute p_{lens} on consensus-A systems barely moves (0.05 → 0.09 — exactly as instructed, since p_{lens} was kept absolute). The framing also has a price that confirms it adds no information: Sonnet under triage rejects only 40% of the random-galaxy negatives that the frozen baseline rejected at 63% —

the rubric shifts each model’s operating point along the same recall/precision curve rather than lifting the curve. Prompting changes what the agent *does* under uncertainty (hedge and hand to a human, rather than reject with confidence); it cannot conjure separability that the 1.3'' pixels do not contain. That is the same wall, now isolated to the data: the fix is better pixels (§10), not a better prompt and not a bigger model.

The three worked-example systems of Appendix A (recorded under the *original* rubric) trace the ablation in miniature. The consensus-A lens is framing-stable (Sonnet B/ $p_{\text{lens}}=0.75$ original vs. B/0.74 triage; Opus C either way). The borderline grade-C system is exactly where the policy bites: Sonnet flips D $\rightarrow$ C with escalation under triage, and Opus, still grading D, now escalates instead of silently discarding. The Grade-D merger is rejected without escalation under all four configurations — a confident rejection of a *nameable* contaminant is robust to framing, which is precisely the behavior the asymmetric rule is meant to protect.

7 Spectroscopic pre-grading

On the gold spectroscopic set the catalog+imaging grader re-grades 20/20 of the 20 Hsu Table-2 Grade-A pairs as plausible lenses and rejects 4/4 of the 4 Foundry-II confirmed non-lenses, using the SIS θ_E /separation consistency and the DR10 imaging.

8 The Foundry-I modelability criterion

The graders so far apply the Huang-2020 discovery rubric (imaging) and Hsu-2025 + physical-consistency reasoning (spectroscopy). The one Foundry-paper criterion runnable on a grz cutout is Foundry-I’s (Huang 2025a): *does a plausible lens MODEL reproduce the configuration?* We implement it as `quick_lensmodel` — a real GIGA-Lens MAP fit (EPL+shear mass, two Sérsic lens-lights, Sérsic+shapelet source) run per cutout in the GIGA-Lens venv (~ 45 s, GPU, cached), keeping JAX out of the SDK process. The discriminator is a `lens_score` combining the recovered θ_E and the χ^2 improvement from ray-tracing the source through the mass (large for a true lens, ~ 0 for a smooth galaxy). The agent can call it as an extra tool (it does: verified end-to-end).

Crucially, evaluating it answers whether a *physical* check beats the vision graders on the hard pool — and it does not. On a balanced validation set the `lens_score` separates real lenses from *ordinary* galaxies well (AUC 0.77) but fails to separate them from the *hard Grade-D human-rejects* (AUC 0.44; the rejects’ mean score is slightly *higher* than the lenses’). A held-out-slice run corroborates the pattern: `lens_score` is flat across graded-B/C, Grade-D rejects, and random galaxies. The interpretation is the same as for vision: Grade-D rejects are CNN-flagged objects that look lens-like enough to admit an EPL fit, so on ground-based *grz* a lens model reproduces them about as well as it reproduces true lenses. **Vision, a stronger backbone, multi-agent deliberation, and a physical lens-model fit all hit the same wall on the hard pool** — strong, convergent evidence that the limit is the data + the soft labels, not the grader. The modelability tool remains useful in its proper regime: filtering real lenses from ordinary galaxies (the broad finder pool), not from already-vetted hard rejects.

9 Engineered representations (RepresentationKit)

The graders so far feed the model the raw Lupton-RGB and let it infer the lensing geometry. We instead built a kit of physically-motivated transforms that make the signal explicit and emit scalar lensing-features (`common/representations.py`; the `lens_representations` tool): improved

lens-light subtraction, a polar/tangential (r, θ) transform (an arc becomes a horizontal bar), a 180° symmetry residual (a counter-image pair lights up), a blue-excess color-isolation map at the Einstein radius, and a DoG+Hessian "arcness" ridge. Each feature is scored as a standalone discriminator (`eval/run_representations.py`) over a larger 545-system manifest with bootstrap CIs, on three contrasts.

Table 4: RepresentationKit standalone feature AUC (best single / logistic combo), with the agent rich-view arm. Consensus-referenced; no human ceiling.

Contrast	best feature	combo
HARD: graded A/B/C vs Grade-D human-reject	0.57	0.51
EASY: graded A vs ordinary galaxy	0.72	0.70
GOLD: Foundry-II confirmed vs non-lens	0.80(tangential/parity)	—
SILVER: noiseless sims, lensed vs unlensed	0.80(arcness/asym.)	—
Agent (48-slice): lean RGB AUC 0.44 vs +representations 0.40		

The kit closes the question cleanly. The features *demonstrably* extract lensing signal: on noiseless sims arcness/asymmetry reach AUC 0.80, and on spectroscopically *confirmed* labels the tangential-extent and counter-image-parity features reach 0.80. They also give a cheap easy-regime pre-screen (combo 0.70, ~ 50 ms vs the 45 s GIGA-Lens fit). But on the *hard consensus pool* they collapse to combo 0.51 — and that does not change with cleaner Tier-2 implementations (photutils isophote, skimage Frangi, SEP geometry: all ≤ 0.57) *nor* when the representations are fed to the agent (0.40 vs lean 0.44, at +43% cost). The *same features* score 0.80 on confirmed labels and 0.51 on the consensus pool — the sharpest evidence in this report that the hard-pool wall is a **label/data limit, not an algorithm limit**: vision, a stronger backbone, deliberation, a GIGA-Lens fit, hand-engineered representations (Tier-1 and Tier-2), and the agent reading those representations *all* land at chance on the soft Grade-D pool, while every method that is tested against clean (confirmed or simulated) labels succeeds.

10 External validation: do better data and more graders break the wall?

The RepresentationKit result localises the imaging wall to the labels and the $\sim 1.3''$ ground-based PSF. We test that directly: crossmatch the candidates against surveys that supply either *higher resolution* or *more graders*, then re-run the grader on the high-resolution pixels (`eval/crossmatch_external.py`, the external-archive workflow, and `eval/run_euclid.py`).

The data exist. Of the 4,354 unique candidates, **291 have external higher-quality data**: 104 in SuGOHI (HSC $0.6''$, a ~ 9 -grader “committee” grade plus, for many, spectroscopic z_l, z_s, θ_E); 148 of the 1,414 grade-A and grade-B candidates have archival *HST* imaging ($0.05\text{--}0.1''$; ACS/WFC, WFC3) in MAST (99/481 grade-A, 49/933 grade-B); 77 are AGEL DR2 Keck-confirmed lenses; and 24 fall in the Euclid Q1 Strong Lensing Discovery Engine field (VIS $0.1''$, ~ 10 expert votes each). **89 of the 291 are DESI grade-C** — precisely the soft pool where the agent and the consensus are unsure.

The “C” grade is information-limited at ground resolution. For the 24 candidates that Euclid independently rediscovered, the DESI $1.3''$ grade and the Euclid $0.1''$ 10-expert grade disagree

in one direction: **53% of the DESI grade-C candidates are graded A or B by the Euclid panel** (6 of 17 jump to grade A). The “C” was not a weak lens; it was an unresolved one.

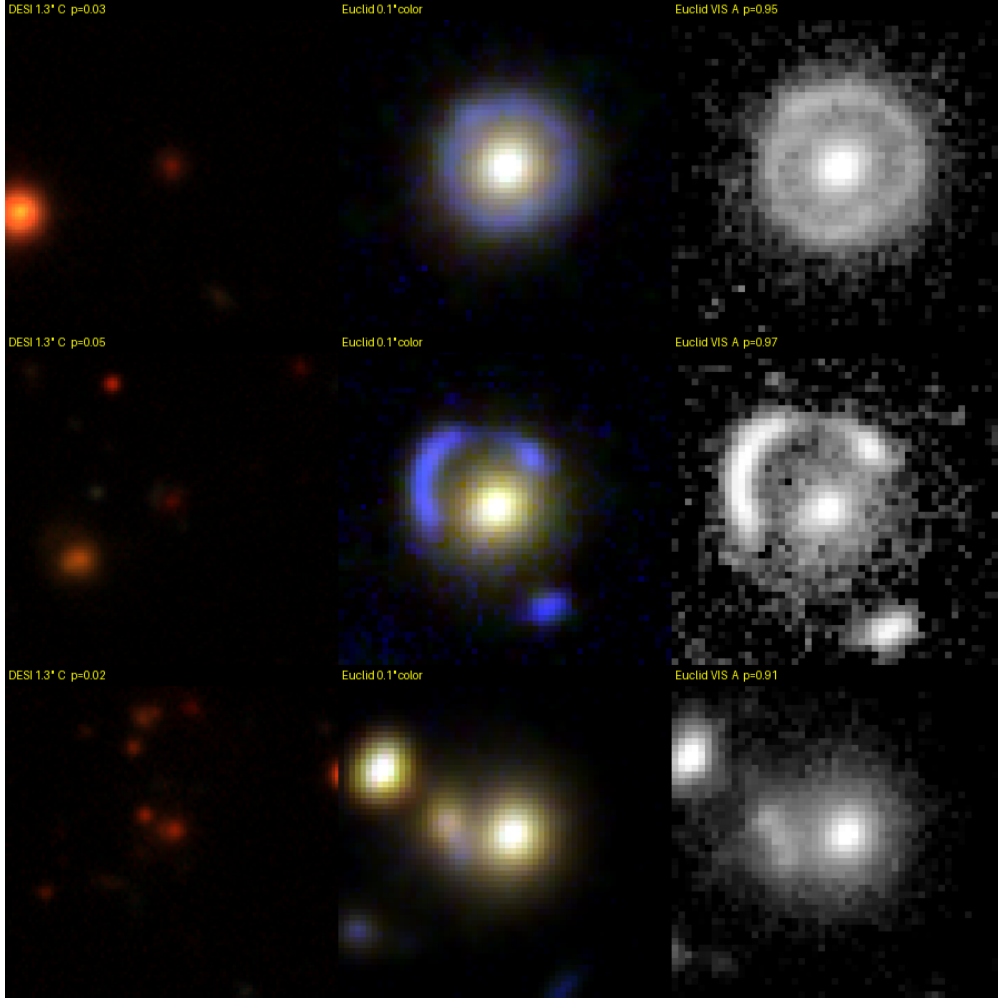


Figure 1: Three DESI grade-C candidates that fall in Euclid Q1. *Left:* DESI grz at 1.3'' seeing (p_{lens} 0.02–0.05; the agent grades D). *Middle/right:* Euclid 0.1'' colour and lens-subtracted VIS — unmistakable Einstein rings/arcs (p_{lens} 0.91–0.97; the agent and 10 Euclid experts grade A). Same agent, same rubric; only the pixels change.

Resolution breaks the *detection* wall (paired, within-object). For the 10 matched candidates with Euclid cutouts on disk we grade the *same object* twice — on the DESI grz cutout and on the Euclid VIS+NIR cutout (`fetch_euclid_cutout`, a 0.1'' analog of `fetch_cutout`). Mean p_{lens} rises from 0.14 (DESI) to 0.75 (Euclid), increasing on 9 of 10; on the DESI grade-C subset it goes from 0.05 to 0.90. The objects the agent rejected as flat blobs at 1.3'' it grades as clear rings at 0.1'' (Fig. 1), agreeing with the Euclid experts. This is the within-object proof that the hard-pool wall was resolution- (and label-) limited, not algorithm-limited.

But the fine A/B/C grade stays soft even at 0.1''. Ranking 95 Euclid candidates by p_{lens} against the 10-expert grade gives Spearman $\rho = 0.08$ (n.s.) and A-vs-C AUC = 0.51 — chance. This is the RepresentationKit finding again at 13× the resolution and 10× the graders: the binary

detection question is resolution-limited and opens up with better data, but the *fine confidence* grade A/B/C is an irreducibly soft human judgment that resolution does not make reproducible. (The grade-C objects with Euclid cutouts are the modelable, lens-like C’s, so this is a conservative contrast.)

11 The human ceiling is low — and now estimated

Every imaging number above is consensus-referenced because the DESI grades are a single 2-author consensus with no per-rater votes. Two independent routes now bound the human ceiling (`eval/human_ceiling.py`); per-classifier raw votes turn out to be unavailable publicly from any survey (Space Warps releases only skill-weighted SWAP posteriors; SuGOHI/Euclid release only aggregated consensus), so we use inter-*team* agreement plus the published literature.

Inter-team agreement (computed, data in hand). The crossmatch gives the same candidates graded independently by other expert groups, so their agreement estimates the ceiling. On a quadratic-weighted κ : DESI team vs the SuGOHI ~ 9 -grader committee = 0.29 (95% CI 0.12–0.44; $n = 103$); DESI team vs the Euclid ~ 10 -expert panel = 0.17 ($n = 24$; partly a genuine resolution effect, §10). Independent expert teams agree only “fairly” on A/B/C.

Published human reliability (literature). This matches the field. Rojas et al. (Rojas et al., 2023) (55 classifiers, 1489 images) find only 73% of repeat gradings identical (intra-rater, an *upper* bound; $\sim$ QWK 0.8) and need ≥ 6 graders to stabilise a marginal grade; Petrillo et al. (Petrillo et al., 2019) report that of candidates flagged by ≥ 1 of 7 inspectors only 4.5% were flagged by all 7; Cañameras et al. (Cañameras et al., 2021) find only 30% of a blind set had zero dispersion among 5 graders; the Euclid Q1 engine (Euclid Collaboration et al., 2025) uses 10 experts/object yet its panel is decisive on only $\sim 38\%$ of candidates. The A/B/C grade is intrinsically soft.

Where the agent sits. On the same SuGOHI-matched set at DESI resolution, LensJudge agrees with the DESI consensus at only $\kappa = 0.00$ ($n = 104$) — *below* the human inter-team ceiling (essentially uncorrelated with the consensus): it is uniformly over-skeptical on $1.3''$ cutouts, grading 66 of these 104 HSC-confirmed candidates “D” and none “A” because the arcs are unresolved. It reaches human-level only when given resolution that disambiguates the lens (the Euclid paired test, §10; $p_{\text{lens}} 0.05 \rightarrow 0.90$). The operative limits, in order: (i) the fine A/B/C grade is barely reproducible by anyone ($\kappa \sim 0.3$ between expert teams, 73% intra-rater); (ii) at ground resolution the agent underperforms even that, over-rejecting blurred lenses; (iii) high-resolution data removes the ambiguity for the binary detection question. The actionable reading, now quantified: deploy LensJudge as a high-recall detect/escalate pre-filter and reserve the “grade” for confirmed or high-resolution data, rather than treating it as a target to reproduce.

12 Caveats and what is not done

(1) **No per-rater human ceiling** — all DESI labels are single consensus. We now anchor a subset against external *multi-grader* grades (Euclid Q1 ~ 10 votes; SuGOHI committee) and HST/Keck confirmation for 291 candidates, but a per-rater κ still needs either the ~ 250 -cutout in-house study or Zooniverse per-classifier votes (consensus is all that is publicly downloadable). (2) **Grade-D negatives are lens-like** (CNN-flagged), so the binary AUC is lower — and more honest — than against random galaxies; strata are scored separately. (3) The **CNN pre-filter gate** (a cost lever

at full-survey scale) is not exercised here because the eval set is by construction the high- p_{meta} ambiguous middle band. (4) Orchestration of the §9.1 specialists is *programmatic* rather than LLM-dispatched **AgentDefinition** subagents, for reliability. (5) Spectroscopic raw-flux streaming is a documented enhancement.

13 Conclusions

LensJudge shows an agentic VI grader can be *built* on the Claude Agent SDK and, more importantly, *tested* on data already in hand. The test returns a nuanced answer. The agent is strong where the task has hard labels and clean evidence — spectroscopic discordant-redshift pairs (20/20 Grade-A plausible, 4/4 non-lenses rejected). On imaging grading of the CNN’s hard high-score pool it is near-chance against the single-consensus A/B/C labels, and — the load-bearing result — *no* stronger backbone or multi-agent design changes that. Because those labels are single-consensus over mostly-unconfirmed candidates, we cannot say whether the agent or the consensus is closer to truth: the multi-grader ceiling is the thing that decides the question — and we now estimate it (§11). Independent expert teams agree only at $\kappa \sim 0.29$ on the A/B/C grade (the literature concurs: 73% intra-rater), so the grade is intrinsically soft; at ground resolution the agent sits *below* even that low ceiling, while high-resolution data closes the gap for the binary detection question. Two concrete next steps follow directly: (i) collect a ~ 250 -cutout, ≥ 2 -grader study to establish the human-vs-human κ and re-anchor every imaging number; (ii) push agentic effort toward the *spectroscopic* channel and toward candidates with confirmation, where the evidence is decisive and the agent already performs. A pre-filter that confidently clears obvious non-lenses while escalating the genuinely ambiguous — which is what LensJudge does — remains useful even at chance agreement on the subjective middle, but its accuracy cannot be claimed until the ceiling exists.

Crucially, both next steps are now *actionable with data we located*. 291 candidates (including 89 of the soft grade-C ones) have external high-resolution imaging, multi-grader grades, or spectroscopic confirmation — from Euclid Q1 (0.1", ~ 10 votes), SuGOHI/HSC, archival HST, and AGEL. Re-grading the same objects on Euclid pixels settles the question for *detection*: p_{lens} on the hard-pool grade-C candidates rises from 0.05 at 1.3" to 0.90 at 0.1", where the agent and 10 experts agree they are clear lenses. The wall the rest of this report documents is therefore the joint product of ground-based resolution and soft single-consensus labels — not a ceiling on the agent. What high resolution does *not* fix is the fine A/B/C confidence grade (A-vs-C AUC 0.51 at 0.1"), confirming that the residual “grade” is a subjective human quantity best replaced, for the agent’s purposes, by a binary detect/escalate decision anchored to confirmed or high-resolution truth.

A The agent in action: three worked examples

This appendix shows the lean single-agent grader (§3; the lean-Sonnet row of Table 1) grading three real LensBench-VI members end-to-end: a consensus Grade-A lens, an ambiguous Grade-C “possible”, and a Grade-D human-rejected contaminant. Episodes were recorded with full message-stream capture (`eval/run_appendix_examples.py`; the benchmark traces log tool events only) using the identical configuration scored in Table 1 — the *original* rubric of §3, not the triage variant of §6.1: `claude-sonnet-4-6` on the Claude Agent SDK, the `fetch_cutout` + `get_photometry` tool pair, a 6-turn limit and a \$0.50/candidate budget cap (observed: \$0.066–\$0.147 and 31–59s per episode). The agent’s *reasoning* is shown in “thinking” boxes at the position it occurred. Two caveats: the displayed text is the API’s *summarized* reasoning, not the raw chain of thought, and

adaptive thinking is the harness default — the instrumented recordings in §6 make visible what earlier runs (including the Table 1 benchmark) performed silently. By design the lean grader is prompted to “respond with ONLY the JSON object”, so beyond the thinking blocks its deliberation surfaces in the tool-call sequence and in the structured **criteria/rationale** fields of its answer. Three editing rules apply: (i) abbreviation is by *elision only*, marked [...] — the agent’s words (thinking included) are never paraphrased; (ii) a harness-internal tool-discovery call at the start of each episode and the `mcp__lens__` tool-namespace prefix are elided, and repeated tool-output boilerplate is shown in the first episode only; (iii) the PNG views the tool returns to the agent are reproduced as figure strips (re-rendered pixel-identically from the same cutout cube). The three episodes were selected for legibility from 9 reasoning-instrumented recordings of the same three systems (none of which happened to invoke `get_photometry` — the Opus 4.8 episode excerpted in §6 shows that exchange); each example cross-references the grade the same system received in the frozen Table 1 benchmark run.

A.1 Positive: a consensus Grade-A lens

DESI-091.7214–58.9787 — consensus grade A, CNN $p_{\text{meta}} = 0.9998$; the benchmark lean run graded it B with $p_{\text{lens}} = 0.75$.

task message

Grade this strong-lens candidate. name='DESI-091.7214-58.9787', survey='storfer'. RA=91.721434, Dec=-58.978734
Tractor type: DEV, region: DECaLS.
CNN ensemble scores (prior, not ground truth): $p_{\text{meta}}=1.000$.

Call `fetch_cutout` with that name and survey, inspect the views, then respond with ONLY the JSON object.

thinking (summarized)

The user wants me to grade a strong-lens candidate. I need to:

1. Call ‘`fetch_cutout`’ to get the views
2. Optionally call ‘`get_photometry`’ and other tools
3. Return a JSON grade

Let me first load the schema for ‘`fetch_cutout`’ and ‘`get_photometry`’ tools.

tool call — `fetch_cutout`

```
{"name": "DESI-091.7214-58.9787", "survey": "storfer", "ra": 91.721434, "dec": -58.978734, "views": ["full", "zoom", "residual"]}
```

tool result — `fetch_cutout` (3 images; Figure 2)

Candidate DESI-091.7214-58.9787 --- grz cutout, 26.5" field. Views follow:
[full] whole cutout, 26.5" across (~0.26"/px), Lupton-RGB (z=R, r=G, g=B). Lens galaxies are red/orange; lensed sources are typically blue.
[zoom] 2.5x center crop --- the 1-5" region where arcs/counter-images appear.
[residual] lens-light removed (per-band minus a smoothed model): low-surface-brightness arcs and counter-images stand out against a flat background.

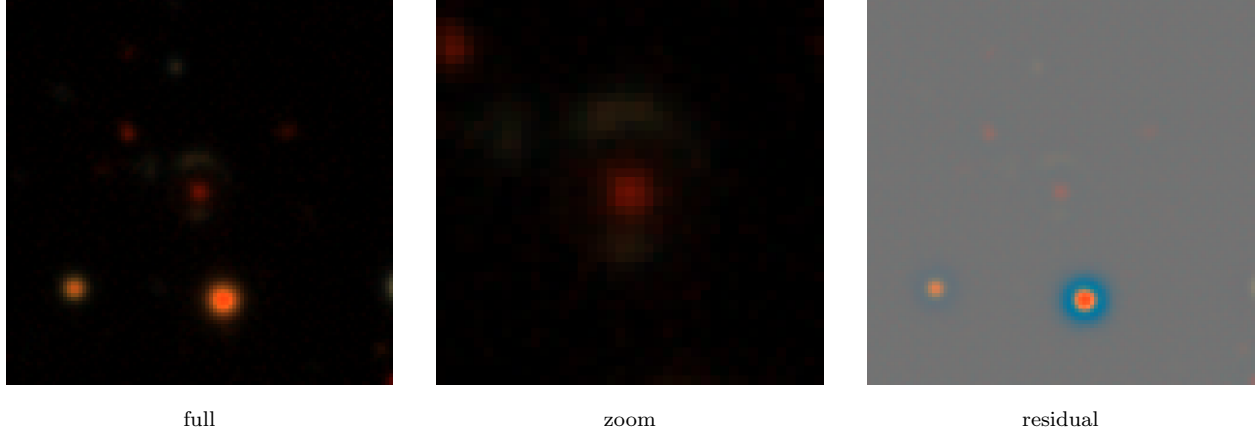


Figure 2: Views the agent inspected for DESI-091.7214–58.9787 (consensus A; this episode: agent grade B, $p_{\text{lens}} = 0.76$). Identical render to the tool output returned to the agent (101 px grz cutout, $0.262''/\text{px}$, Lupton RGB).

thinking (summarized) — after inspecting the views

Let me analyze the three views carefully:

****Full view (26.5" field):****

- The field shows a central galaxy that appears reddish/orange, surrounded by several other galaxies
- There's a slightly blue/whitish feature visible near the central red galaxy --- in the upper-left area of the central object

[...]

****Zoom view (center crop, 1-5" region):****

- [...] There's a blue/cyan feature to the upper-left of the red core that could be a lensed arc or blue companion galaxy, appearing somewhat diffuse without an obvious counter-image on the opposite side

****Residual view (lens light removed):****

- The residual image reveals a striking blue ring or halo structure surrounding a bright central source
[...] --- this is strong evidence for gravitational lensing [...]
- The lens-light removal has exposed this blue ring/arc that was harder to see in the full view

[...]

Now I'm evaluating this against the five lensing criteria. The blue source is clearly present in both the zoom and residual views, so that scores around 7-8. [...] Actually, let me reconsider the grading criteria more carefully. For an A grade, I need strong evidence from at least two of curvature, counter_images, or arc_morphology, plus no decisive contaminants. Looking at what the residual shows: the blue emission curves around the central red galaxy (strong curvature evidence), and it's clearly extended and ring-like rather than point-like (strong arc morphology evidence). A ring structure would imply counter-images exist, giving moderate support there too. [...] This could genuinely be an A grade, though the noisy imaging makes me hesitant to commit fully. [...] Finalizing this as a ****B**** grade with moderate confidence around 0.65 and a lens probability of 0.75-0.80, escalating to human review given the ambiguity about whether this is a complete symmetric ring or just a nearby blue companion.

agent — final grade (JSON)

```
{
  "grade": "B",
  "criteria": {"blue_source": 7, "low_surface_brightness": 6, "curvature": 7,
    "counter_images": 6, "arc_morphology": 8},
  "p_lens": 0.76,
  "confidence": 0.63,
  "contaminant": null,
  "escalate_to_human": true,
  "rationale": "The residual view is the key diagnostic: after lens-light subtraction a clear cyan/blue extended halo appears encircling the central orange-red DEV galaxy, consistent with a partial-to-
```

```

complete Einstein ring. The zoom view corroborates a blue feature offset ~1--2" to the upper-left of
the red core (criterion 1 and 3). Arc morphology is the strongest signal --- the blue emission is
diffuse and elongated rather than point-like, and its curvature tracks the centre. Counter-images
cannot be cleanly resolved into distinct opposing arcs at this resolution, preventing an A; the ring-
like envelope in the residual is suggestive but ambiguous between a near-complete ring and an LSB blue
companion. No spiral, stellar halo, or tidal-tail contaminant is apparent. Escalated for human review
given the A/B borderline."
}

```

The reasoning narrates what the structured output alone cannot: a per-view visual analysis, criterion-by-criterion scoring, and an explicit A-versus-B deliberation that ends in escalation. Two further reasoning-instrumented recordings of the same system returned B ($p_{\text{lens}} = 0.72$) and B ($p_{\text{lens}} = 0.62$) — stable at B, bracketing the frozen benchmark run’s B/0.75.

A.2 Neutral: an ambiguous Grade-C “possible”

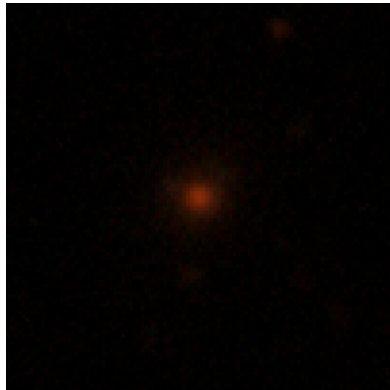
DESI-030.9742+12.4683 — consensus grade C, CNN $p_{\text{meta}} = 0.898$; the benchmark lean run also graded it C ($p_{\text{lens}} = 0.12$). The task message is analogous to A.1.

tool call — fetch_cutout

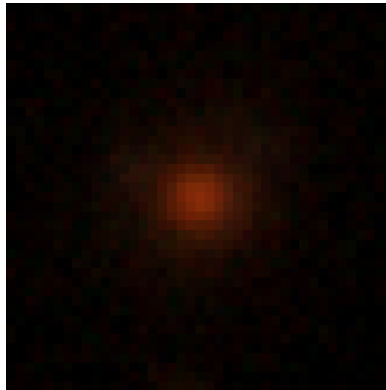
```
{"name": "DESI-030.9742+12.4683", "survey": "storfer", "views": ["full", "zoom", "residual"]}
```

tool result — fetch_cutout (3 images; Figure 3)

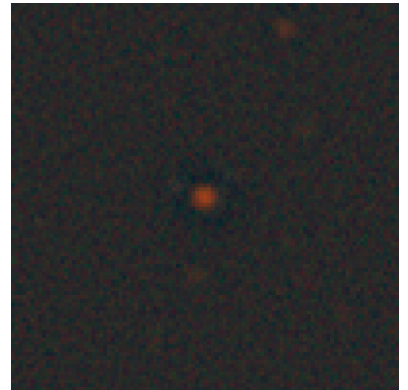
Candidate DESI-030.9742+12.4683 --- grz cutout, 26.5" field. Views follow:
[full] [...] [zoom] [...] [residual] [...]



full



zoom



residual

Figure 3: Views the agent inspected for DESI-030.9742+12.4683 (consensus C; this episode: agent grade C, $p_{\text{lens}} = 0.12$, escalated). A smooth red elliptical with a faint annular excess visible only in the residual.

thinking (summarized) — after inspecting the views

Let me carefully analyze the three views:

****Full view:****

- Central red/orange elliptical galaxy dominating the field
- No obvious blue arcs, rings, or counter-images visible around the galaxy
- [...]

```

**Residual view:**
After subtracting the lens model, I can see the central core remaining---likely indicating incomplete subtraction or a ring structure---but the faint features that emerge are reddish/orange rather than blue. [...]

**Visual assessment against lensing criteria:**
The blue source is essentially absent---I don't see any clear blue feature within 1-5" of the central red galaxy, and the visible residual features are all reddish/orange. [...] The CNN score is quite high at 0.898, but visually this looks like an isolated red elliptical galaxy with possibly some ring-like residual that could just be a PSF or model subtraction artifact rather than a genuine lensing signature.

Looking more closely at the residual, there's a faint ring or partial ring structure around the core, but the color is distinctly orange/reddish rather than blue. This could be a very faint Einstein ring or just an imperfect model subtraction. Given the lack of clearly blue features, no tangential arc, and the residual showing mostly orange/red tones, this is most likely a Grade D or at best Grade C---the ring might hint at some lensing, but it's not convincingly blue. [...] The high CNN score creates some tension with what I'm seeing visually, which makes this a borderline case worth flagging for human review. [...] I'm settling on C with a  $p_{\text{lens}}$  around 0.18 and escalating to human review, though the isolated red elliptical explanation seems most plausible given the ambiguous residual features.

```

agent — final grade (JSON)

```

{
  "grade": "C",
  "criteria": {"blue_source": 2, "low_surface_brightness": 3, "curvature": 1,
    "counter_images": 1, "arc_morphology": 1},
  "p_lens": 0.12,
  "confidence": 0.62,
  "contaminant": null,
  "escalate_to_human": true,
  "rationale": "All three views show a smooth, isolated red elliptical (consistent with the SER Tractor type) with no convincingly blue feature at 1--5\" separation in either the full or zoom frames. The residual reveals a faint ring-like halo around the subtracted core, but its color is orange-red rather than blue, suggesting an imperfect lens-light model fit rather than a lensed source; no tangential curvature or arc morphology is identifiable. The single nearby extended object visible to the upper-right in the residual is also red/orange, ruling it out as a lensed source. Despite the high CNN prior ( $p_{\text{meta}}=0.898$ ), the absence of any blue arc, counter-image, or tangential elongation reduces  $p_{\text{lens}}$  substantially; flagged for human review given the discrepancy between the ML score and the visual evidence."
}

```

The reasoning shows the C/D boundary being weighed in real time — “most likely a Grade D or at best Grade C” — before settling on the escalated “possible”. Two further recordings of the same system landed on the other side of that line (D, $p_{\text{lens}} = 0.07$ and D, $p_{\text{lens}} = 0.12$): the same evidence reads as “possible” or “rejected” on different runs, which is precisely the soft boundary §11 quantifies.

A.3 Negative: a Grade-D human-rejected contaminant

DESI-124.2632+17.1177 — consensus grade D (human-rejected), yet CNN $p_{\text{meta}} = 0.91$; the benchmark lean run graded it C ($p_{\text{lens}} = 0.32$). The task message is analogous to A.1.

tool call — fetch_cutout

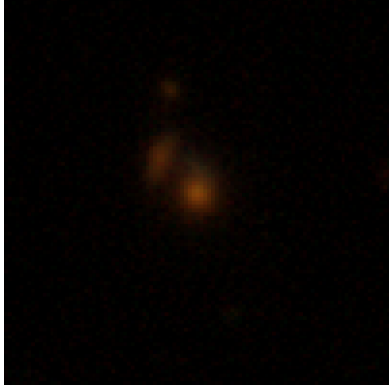
```

{"name": "DESI-124.2632+17.1177", "survey": "storfer", "views": ["full", "zoom", "residual"], "ra": 124.263204, "dec": 17.117705}

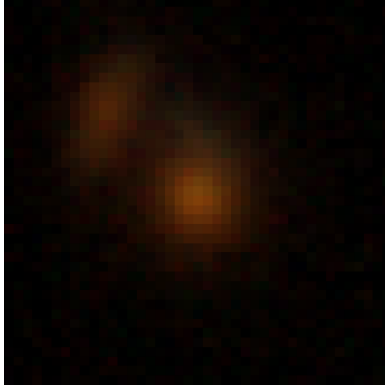
```


tool result — fetch_cutout (3 images; Figure 4)

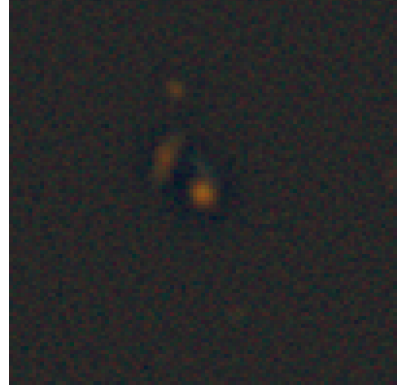
Candidate DESI-124.2632+17.1177 --- grz cutout, 26.5" field. Views follow:
[full] [...] [zoom] [...] [residual] [...]



full



zoom



residual

Figure 4: Views the agent inspected for DESI-124.2632+17.1177 (consensus D; this episode: agent grade D, $p_{\text{lens}} = 0.05$, contaminant merger). A close pair of red galaxies that the CNN scored $p_{\text{meta}} = 0.91$.

thinking (summarized) — after inspecting the views

Let me analyze the three views I received:

****Full view:****

- There's a central red/orange galaxy (the potential lens)
 - There appears to be another orange/red blob to the upper-left of the main galaxy
 - No obvious blue features are visible - both features appear orange/red in color
- [...]

Looking at all three views together, this system appears to be dominated by two red/orange galaxies in close proximity with no evidence of blue lensed features, arcs, or a background source - it's more consistent with a galaxy group or interaction rather than a strong lensing system. The CNN flagged this with a high score of 0.910, but that's a classic false positive---red galaxy pairs can easily trigger the model without any actual lensed blue source present. [...] I'm confident enough in the D grade and marking this as not needing human escalation.

agent — final grade (JSON)

```
{
  "grade": "D",
  "criteria": {"blue_source": 1, "low_surface_brightness": 2, "curvature": 0,
    "counter_images": 0, "arc_morphology": 0},
  "p_lens": 0.05,
  "confidence": 0.82,
  "contaminant": "merger",
  "escalate_to_human": false,
  "rationale": "All three views reveal a red/orange galaxy pair in close angular proximity (~2--3\") with no blue source, arc, or tangential feature at any contrast level. The zoom confirms both components are similarly red/orange --- consistent with two early-type galaxies at similar redshifts or a galaxy group rather than a lens+background-source configuration. The residual view shows only orange over-subtraction artifacts and a faint reddish satellite; no blue LSB emission, curvature, or counter-image is detected. The high CNN score (p_meta=0.910) is a false positive driven by the compact red galaxy pair morphology."
}
```

The shortest reasoning of the three episodes (2,051 summarized characters versus 4,220 for A.1): the contaminant is recognized at a glance — “that’s a classic false positive — red galaxy pairs can easily trigger the model” — and the agent declines to escalate. A second recording reproduced $D/p_{\text{lens}} = 0.05$ exactly. The confident rejection with a named contaminant is the pre-filter behavior the report argues is the agent’s useful role on ground-based imaging.

References

- R. Cañameras, S. Schuldt, S. H. Suyu, et al. HOLISMOKES VI: New galaxy-scale strong lens candidates from HSC. *Astronomy & Astrophysics*, 653:L6, 2021. doi: 10.1051/0004-6361/202141758.
- Euclid Collaboration, M. Walmsley, P. Holloway, et al. Euclid Quick Data Release (Q1): The Strong Lensing Discovery Engine A – System overview. *arXiv e-prints*, 2025.
- B. Hsu, X. Huang, et al. A Pair-wise Spectroscopic Search for Strong Gravitational Lenses in DESI DR1. *arXiv e-prints*, 2025.
- X. Huang, C. Storfer, V. Ravi, et al. Finding Strong Gravitational Lenses in the DESI DECam Legacy Survey. *The Astrophysical Journal*, 894(1):78, 2020. doi: 10.3847/1538-4357/ab7ffb.
- X. Huang, C. Storfer, A. Gu, et al. Discovering New Strong Gravitational Lenses in the DESI Legacy Imaging Surveys. *The Astrophysical Journal*, 909(1):27, 2021. doi: 10.3847/1538-4357/abd62b.
- X. Huang et al. DESI Strong Lens Foundry II: DESI Spectroscopy. *arXiv e-prints*, 2025.
- J. Inchausti Reyes, C. Storfer, X. Huang, et al. A Search for Strong Gravitational Lenses in DESI-DR10 with Two Neural-Network Architectures. *arXiv e-prints*, 2025.
- C. E. Petrillo, C. Tortora, G. Vernardos, et al. LinKS: discovering galaxy-scale strong lenses in KiDS using CNNs. *Monthly Notices of the Royal Astronomical Society*, 484(3):3879–3896, 2019. doi: 10.1093/mnras/stz189.
- K. Rojas, T. E. Collett, D. Ballard, et al. The impact of human expert visual inspection on the discovery of strong gravitational lenses. *Monthly Notices of the Royal Astronomical Society*, 523(3):4413–4430, 2023. doi: 10.1093/mnras/stad1680.
- C. Storfer, X. Huang, et al. New Strong Gravitational Lenses from the DESI Legacy Imaging Surveys DR9. *The Astrophysical Journal Supplement Series*, 274(2):16, 2024. doi: 10.3847/1538-4365/ad646e.